paragraph 2.5 of the EEDI Calculation Guidelines; and

.2 for ships for which sea trial cannot be conducte d under the

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SOURCE: MEPC.261(68).pdf
speed at sea trial at the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines; and

.2 for ships for which sea trial cannot be conducte d under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines: if the measured ship speed at the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines at the sea trial conditions is different from the expected ship speed on the power curve at the corresponding condition, the shipbuilder should recalculate the attained EEDI by adjusting ship speed under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines by an appropriate correction method that is agreed by the verifier.

RESOLUTION MEPC.261(68)
(adopted on 15 May 2015)
AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION OF THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)
(RESOLUTION MEPC.254(67))MEPC 68/21/Add.1
Annex 6, page 4

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
An example of scheme of conversion from trial condition to EEDI condition at EEDI power is given as follows:

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SOURCE: MEPC.261(68).pdf
(RESOLUTION MEPC.254(67))MEPC 68/21/Add.1
Annex 6, page 4

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
An example of scheme of conversion from trial condition to EEDI condition at EEDI power is given as follows:

Vref is obtained from the results of the sea trials at trial condition using the speed-power curves predicted by the tank tests. The tank tests shall be carried out at both draughts: trial condition corresponding to that of the S/P trials and EEDI condition. For trial conditions the power ratio P between model test prediction and sea trial result is calculated for constant ship speed. Ship speed from model test prediction for EEDI condition at EEDI power multiplied with P is Vref.
STrial
PTrial
P
P

P
,
,

where:
PTrialP ,
: power at trial condition predicted by the tank tests
STrialP ,
: power at trial condition obtained by the S/P trials
P
: power ratio

Figure 2 shows an example of scheme of the conversion to derive the resulting ship speed at EEDI condition (
refV) at EEDI power.

Figure 2: An example of scheme of conversion from trial condition to EEDI condition at EEDI power

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SOURCE: MEPC.261(68).pdf
Figure 2 shows an example of scheme of the conversion to derive the resulting ship speed at EEDI condition (
refV) at EEDI power.

Figure 2: An example of scheme of conversion from trial condition to EEDI condition at EEDI power

Note: Further consideration would be necessary for speed adjustment methodology in paragraph 4.3.9.2 of these guidelines. One of the concerns relates to a possible situation where the power curve for sea trial condition is estimated in an excessively conservative manner (i.e. power curve is shifted in a leftward direction) with the intention to get an upward adjustment of the ship speed by making the measured ship speed at sea trial easily exceed the lower-estimated speed for sea trial condition at design stage."

***
MCR
NCR
EEDI power
Power
EEDI condition Trial condition
Tank test Results
Sea Trial Results
Adjusted speed by the
Results of Sea trial

refV

Ship Speed
P * EEDI power
RESOLUTION MEPC.261(68)
(adopted on 15 May 2015)
AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION
OF THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)
(RESOLUTION MEPC.254(67))RESOLUTION MEPC.261(68)
(adopted on 15 May 2015)

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SOURCE: MEPC.261(68).pdf
RESOLUTION MEPC.261(68)
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AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION
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(RESOLUTION MEPC.254(67))RESOLUTION MEPC.261(68)
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AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION
OF THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)
(RESOLUTION MEPC.254(67))

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SOURCE: MEPC.262(68).pdf
RESOLUTION MEPC.262(68)
(adopted on 15 May 2015)
AMENDMENTS TO THE 2013 INTERIM GUIDELINES FOR DETERMINING MINIMUM PROPULSION POWER TO MAINTAIN THE
MANOEUVRABILITY OF SHIPS IN ADVERSE CONDITIONS
(RESOLUTION MEPC.232(65), AS AMENDED BY
RESOLUTION MEPC.255(67))MEPC 68/21/Add.1
Annex 7, page 1

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
ANNEX 7

RESOLUTION MEPC.262(68)
(adopted on 15 May 2015)

AMENDMENTS TO THE 2013 INTERIM GUIDELINES FOR
DETERMINING MINIMUM PROPULSION POWER TO MAINTAIN THE
MANOEUVRABILITY OF SHIPS IN ADVERSE CONDITIONS
(RESOLUTION MEPC.232(65), AS AMENDED BY RESOLUTION MEPC.255(67))

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,

RECALLING ALSO that, at its sixty-second session, it adopted, by resolution MEPC.203(62),

Amendments to the annex of the Protocol of 1997 to amend the International Convention for

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SOURCE: MEPC.262(68).pdf
RECALLING ALSO that, at its sixty-second session, it adopted, by resolution MEPC.203(62), Amendments to the annex of the Protocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that the aforementioned amendments to MARPOL Annex VI entered into force on 1 January 2013,

NOTING ALSO that regulation 21.5 of MARPOL Annex VI, as amended, requires that the installed propulsion power shall not be less than the propulsion power needed to maintain the manoeuvrability of the ship under adverse conditions as defined in guidelines to be developed by the Organization,

NOTING FURTHER that, at its sixty-fifth session, it adopted, by resolution MEPC.232(65), the 2013 Interim guidelines for determining minimum propulsion power to maintain the manoeuvrability of ships in adverse conditions (the interim guidelines) and, at its sixty-seventh session, by resolution MEPC.255(67), amendments thereto,

RECOGNIZING that the amendments to MARPOL Annex VI require the adoption of relevant